

Two devices, one family

Tangerine admin desk unit (left) and basalt check-in station (right)



Admin desk unit

110 × 90 × 55.6 mm · sloped 18° · ESP32 + PN532



Check-in station

100 × 76 × 43 mm · ESP32 + RC522 · food / place / activity



Staff register a band on the tray

Charm sits in the Ø50 tray; the lip stops it sliding off



Printable parts

Part	Material	Size (mm)
Admin shell	tangerine PLA/PETG	110 × 90 × 53.2
Admin bottom plate	basalt grey PLA/PETG	110 × 90 × 2.4 (+ cradle)
Station base	basalt grey PLA/PETG	100 × 76 × 40
Station lid	cream PLA/PETG	100 × 76 × 3 (+ skirt)

Both: ESP32 DevKitC, 12 mm buzzer (GPIO 4), 5 mm LED (GPIO 2), 4 × M3 screws, rubber feet.
Admin reads through 1.2 mm at the tray; station reads through 2.2 mm at the tap circle.
The station frame also takes a PN532, so one station print can stand in as an admin unit.
Files: stl/ (print-ready), drawings/ (A3 dimensioned PDF), source/ (parametric Python).